

JORDAN RIVER MINES LTD.

SUNRO MINE

Annual Geological Report

1972

802181

G. Meusy

JORDAN RIVER MINES Ltd

SUNRO MINE

R. V. KERN

ANNUAL GEOLOGICAL REPORT

1972

G. MEUSY

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INTRODUCTION

This report was written to summarize the observations made at the Sunro Mine from January 1, 1972 to December 28, 1972.

In Part One is presented a new rock types classification which is now in use at the mine. This classification is the result of a two year experience in mapping at Sunro. Although not perfect, since it is not backed up by microscopic studies, this rock classification has been found very convenient to map and correlate the workings.

Part Two of the Report explains the zoning observed in the Orebodies and the genetic theory established according to this zoning. The Ore Reserves are presented in this Part Two of the Report.

Part Three gives a summary of the Exploration done in 1972 and proposals for 1973.

Technical Notes on Diamond Drilling and Sampling are given in Part Four.

This report is to be accompanied by the following maps:

- 52 Level 1" = 100'
- 53 Level 1" = 100'
- Long Section along the main adit 1" = 400'

RECOMMENDATIONS

The Total Reserves of Proven and Probable Ore calculated at the end of 1972 amount to 1,666,296 Tons @ 1.46 (Undiluted). Reserves in the same category at the end of 1971 were 1,706,124 Tons @ 1.25. Since 126,778 Tons @ 1.007 were mined out in the year, the increase in tonnage is 86,950 Tons.

A total of 20,693 feet of diamond drilling was required to find and outline this new ore, giving a ratio of 4.2 Tons per foot of drill hole.

Very little short range exploration was done during the year, and no long range exploration was started at all.

The theory about the genetic similarity of the ore zones and their relation with the gabbro intrusive has influenced the exploration proposals for 1973; however, this theory being only recently developed, other exploration proposals have been studied as well.

The target of the long range exploration proposals is to indicate 1,350,000 Tons of new ore, grading approximately 1.20% Copper. The total cost for this exploration is estimated at \$245,000. This exploration is strongly recommended and should start as soon as possible this year.

To take care of the exploration program and to carry out more detailed studies on the petrography and the mineralogy of the deposits, as well as more surface work, more manpower will be required for the Geology Department in 1973.

Respectfully Submitted,



G. Meusy
Mine Geologist

PART ONE

GENERAL GEOLOGY

ROCK TYPES DESCRIPTION

The rock formations described in 1971's Report were only part of the facies recognized to date. New petrographic types were identified this year and a first attempt of systematic classification has been worked out.

However, no microscopic study has been done to date, and the following descriptions are based only on magascopic examination with the help of 10 power magnifying glass.

I - VOLCANICS

I - A - BASALT

Under this name are grouped aphanitic to cryptocrystalline rocks; they look quite massive to the unaided eye and have a dark greyish to greenish colour. Hardness varies from 5.5 to 6.5 on MOHS Scale. The magnifying glass shows a very fine grained texture of white feldspars and dark hornblende crystals; the average size of the crystals must therefore be around 0.2mm. The break of these rocks is always sharp.

I - B - AMYGDALOIDAL BASALT

FIG. I - B

This basalt variety shows a variable percentage of white feldspars amygdules (5% to 40%) in an aphanitic background mass similar to the basalt described in I-A. On several specimens there appears to be a grading in the distribution of these amygdules ("granoclasement") and also a lineation given by the stretching of these amygdules. This could give information about the flow attitude. The size of these amygdules varies from .5 to 5mm.

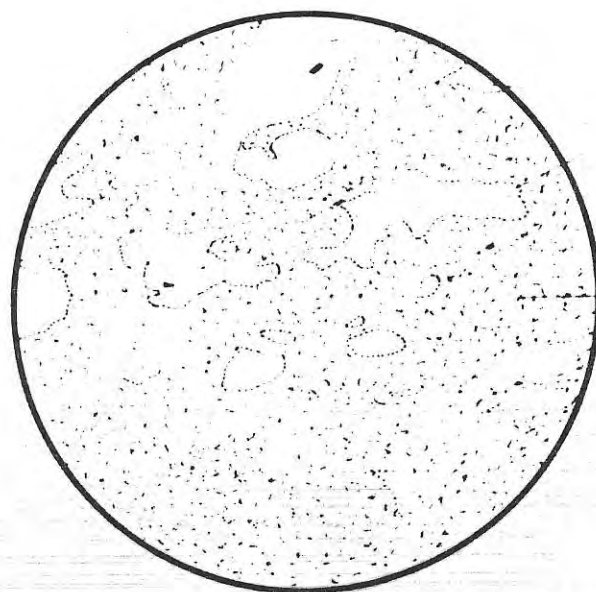
I - C - PORPHYRITIC BASALT

This facies has been mentioned by Stevenson in 1950. However, only one type of porphyritic basalt has been observed by the writer, in which only Hornblende crystals are present.

I - D - DIABASE

FIG. I - D

Megascopically, this rock displays a very characteristic texture where feldspars laths are conspicuous. This texture is very similar to the doleritic texture, but with a background mass more difficult to recognize. A few black crystals are also present, with a stout habitus; they might be Augite.

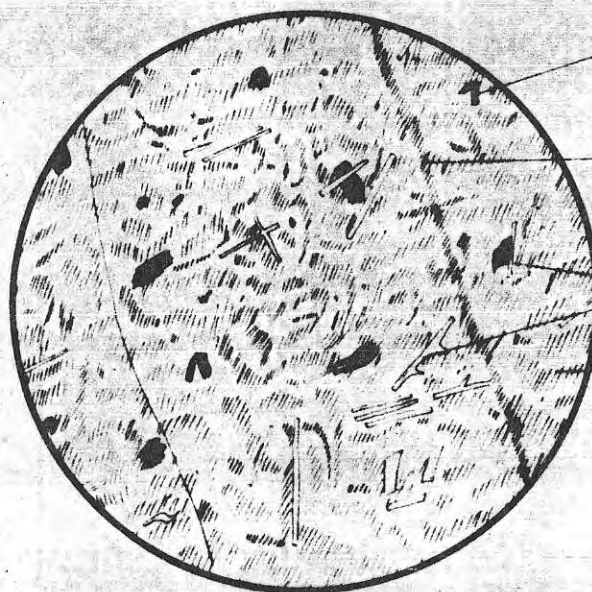


Plagioclase Amygdules

Plagioclase Hornblende

Aphanitic Matrix

FIG I-B AMYGDALOIDAL BASALT (X 10)



Black Crystals
(Hornblende or Augite)

Fissure Filled with
greenish hornblende

Feldspars Laths

Background of "cloudy"
Hornblende and feldspars

FIG I-D DIABASE

(X 10)

This diabase forms masses of very irregular shape and size in the volcanics; however, some diabase masses are also found in the gabbro, where their contact is very sharp; when the diabase occur in the basalt, there is no contact to be observed, but rather a wide transition zone where the grain of the rock gradually decreases.

II - INTRUSIVES

II - A - GABBRO

FIG. II - A

The Gabbro occurring at Sunro is a coarse grained Hornblende Gabbro. Study of hand specimens shows an imbrication of big Hornblende crystals (2 mm to 1 cm and bigger in the pegmatitic type) and large Plagioclase interstitial masses. Two types of Hornblende can be observed, one pale green and the other dark green. The Hornblende crystals are not always euhedral; when they are, the stout shape of the crystals suggests that they were originally Pyroxenes.

Some black crystals are also present and might be remnants of the original Augite. The feldspars are corroded and cut by numerous minute cracks filled with Hornblende or siliceous material.

The rock is very much softer than the volcanics (5.5); the grain varies considerably from a few millimeters to several centimeters in the pegmatitic varieties.

II - B - HORNBLENDITE

FIG. II - B

This rock shows a felted mass of long-bladed Hornblende crystals, up to 2 cm long, but generally around 1 cm long. This Hornblende is fibrous and black. It is cut in places by sulphides veinlets. A few interstitial feldspar grains are sometimes present.

The grain of the rock becomes very fine in the heavily fractured and sheared areas, and can be aphanitic, therefore very similar to some basic dykes or some basalt varieties. In "Gabbro of East Sooke and Rocky Point", Cooke states:

"The faulting has granulated the large-bladed Hornblende crystals, so that much of the Hornblende of these zones is now no more than cemented dust, and has produced numerous slickensided cracks and fissures, while at the same time much of the Hornblende has been converted into Chlorite."

This description of the East Sooke Hornblendite Zones applies perfectly to those mapped in the Sunro Mine.

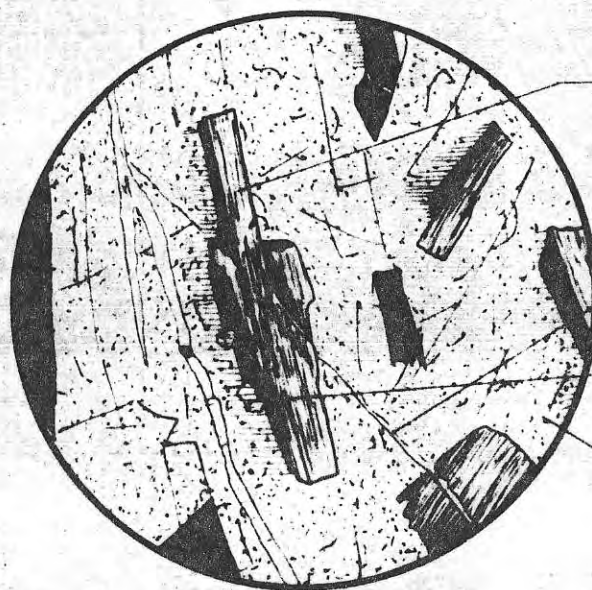
This hornblendite occurs at Sunro either in large zones within the Gabbro, or in veinlets a few inches wide in the Gabbro or in the



- Crystalline Cleavage
- Chalcopyrite Specks
- Plagioclase
- Dark green hornblende
- Pale green hornblende
- Hornblende Crystal
- Black crystals ?
- Augite remnants

FIG II-A GABBRO

(X 10)



- Hornblende Crystal
- Sulphides Veinlet
- Feldspar Grain
- Hornblende Dust
- Background Mass

FIG II-B HORNBLENDITE

(X 10)

Volcanics as well. The crystals are disposed at right angle with the strike of the veinlet and the center part of this veinlet commonly shows sulphides.

III - DYKES

A good description of both acidic and basic dykes has been given in last year's report. It is fully established by now that the basic dykes are always barren and cut the ore zones. The relationship with the acidic dykes is not established yet.

RELATIONS BETWEEN ROCK TYPES

The amygdaloidal and prophyritic basalt varieties form irregular masses in the massive basalt and do not have any sharp contact; their continuity is very doubtful.

The diabase form irregular "puffs" along the mineralized faults of Type II (See "Faults Classification" in 1971 Report) and also seems to favour the proximity of the Gabbro contact. There again, the contact is not sharp; it is rather a broad transition zone where the rock gradually grades into the massive basalt. However, the diabase also occurs within the Gabbro as inclusions, and the contact with the Gabbro is then very sharp, although not faulted. These inclusions were referred to as "Fine grained Hornblendite Inclusions in Gabbro" in 1971 Report.

The contact gabbro-basalt seems to be rather sharp, but several inclusions of diabase located along this contact give a misleading aspect of a broad transition zone. There is no breccia on this contact.

The Hornblendite occurs either in small veinlets cutting the gabbro or the basalt, or in large zones within the gabbro. The small veinlets are only a few inches wide and are composed of a string of Hornblende crystals elongated at right angle with the strike of the veinlet. In the large hornblendite zones, the contacts are gradual, by a progressive increase of the fresh feldspars content; erratic "clouds" and lenses of Hornblendite beside the main mass are quite common.

Both acidic and basic dykes cut all the other rock types, with a very sharp contact and chilled edges.

ROCK TYPES CLASSIFICATION (Regional and at the Mine).

REGIONAL SERIES	GROUP	SUNRO SERIES	AGE
Shales Sandstones Conglomerates	SOOKE SEDIMENTARY FORMATION	Sandstones Conglomerates	MIOCENE
Acidic Dykes Basic Dykes Hornblendites Aplite Veins Granite Anorthosite Hornblende Gabbro Augite Gabbro olivine Gabbro	GABBRO SOOKE	Acidic Dykes Basic Dykes Hornblendites Aplite Veins? Hornblende Gabbro	OLIGOCENE
Diabase Andesites Tuffs Agglomerates Pillow-Lavas Massive Basalts Porphyritic Basalt Amygdaloidal Basalt	METCHOSIN VOLCANICS	Diabase Massive Basalts Porphyritic Basalt Amygdaloidal Basalt	UPPER EOCENE

THEORETICAL PART

I - Relative Age of Rock Types

The oldest rocks in the Jordan River - Sooke area are the Metchosin Volcanics. They show both sub-marine facies (Pillow Lavas) and explosive aerial facies (Agglomerates Pipe known at Albert Head).

These volcanics were probably deposited by several flows and therefore display both textural and mineralogical varieties. These different phases are represented by the amygdaloidal and porphyritic varieties. The amygdaloidal type is probably the result of a pumiceous rock where re-crystallization filled the pores to build amygdules.

As was pointed out to the writer, it is probable that movement took place prior to consolidation in the volcanics, since several "collapse structures" and "slumping structures" have been observed.

The next younger rock type seems to be the Sooke Gabbro, in its un-altered, Augite facies. The Gabbro intruded the volcanics in a most likely very slow process, since no breccia is found on the contact. Maybe the intrusion took place in the weakest layers of the volcanics flow, thus giving sill-like bodies. Consideration should be given to the idea that this intrusion might have taken place when the volcanics were not quite solidified.

The diabase occurring in the volcanics is considered at the present time as a re-crystallization of the basalt, induced by a contact-metamorphism phenomenon. However, the diabase "puffs" along the Type II Faults, away from the intrusive, are of unknown origin.

As far as the diabase inclusions in the Gabbro are concerned, they are considered as masses of volcanics included in the intrusive, metamorphosed by heat and probably brought up with the magma.

The Hornblendite zones are believed to be the result of a replacement of the Gabbro, as they have been proven to be at East Sooke. The theory about these Hornblendite zones developed by H.C. Cooke in "Gabbros of East Sooke and Rocky Point" fits the information collected at Sunro.

According to this theory, these hornblendite zones are "...replacement veins formed by the action on the gabbro of solutions rich in iron and magnesia rising through joint cracks and other fissures." Since the hornblendite veins also occur in the volcanics, it is probable that the solutions also invaded the rock beyond the intrusive, but with a very short extension.

Cooke also states:

"The very large hornblendite zones are identical with the smaller in composition, but differ from them

in two important respects. They have been affected by faulting movements, and the copper ores have been deposited within them."

In the writer's opinion, the small hornblendite veins are only the apical part of the larger hornblendite zones.

Since the lateral extent of the hornblendite is very limited in the veins, the writer is in agreement with Cooke when he writes that "the large zones represent pre-existent, sheeted fault zones which afforded the solutions a multitude of closely spaced planes of slip through which to rise, and thus gave them power to alter broad bands of country rock."

To summarize, the hornblendite zones are typical of a metasomatic replacement of the gabbro, and secondary fracture-filling in the basalt.

As we have seen before, all the above rock types are cut by younger acidic and basic dykes.

Finally, a thin sedimentary cover of sandstones and conglomerates overlays the crystalline group.

II - Summary

The geological history of the district can be summarized as follows:

Eocene: Shallow submarine volcanic activity, resulting in the deposit of approximately 7500 feet of basic rocks, covering the southern tip of Vancouver Island.

Upper Eocene: Creation of N.S. stresses on this volcanic cover, relieved along the Type I and Type II Faults, and also pushing the Leech River Group rocks over the volcanics (Leech River Thrust Fault). These stresses also put additional pressure on the magma reservoir from which the volcanics were issued, originating the intrusive activity.

Lower Oligocene: Intrusive activity resulting in the creation of the Gabbro stocks and sills. Invasion and metasomatic replacement of the gabbro by hot solutions rising through Type I and II Fault Zones, and then, copper-iron ore deposition in the faults and in the hornblendite zones. These solutions probably originated from the cooling magma.

Middle Oligocene: Late tectonic activity creating the Type III Faults, creating also additional movements along the already mineralized Type II Faults. The swarm of dykes was probably put in place at this time, too.

Early Miocene: Coastal sedimentary period.

TECTONICS

The data collected in 1972 does not add much to what has been exposed in last year's report.

The faults classification outlined in 1971 worked well to correlate the new accidents observed this year.

Again, no evidence of folding has been found, although some of the diabase masses penetrated by diamond drill holes show some lineaments due to bands of fine and coarser grain.

PLATE I : FAULTS CLASSIFICATION

Regional Type		Sunro Equivalent		Characteristic Features					
N°	Strike & Dip	Name	Strike & Dip	Chalcopyrite Pyrrhotite	Pyrite	Calcite	Gouge	Dykes	Hornblende
I	N 95-65 E, Vert.	Cross Fault	N 60 E, Vert	YES	NO	NO	YES	NO	?
		Coalfield ??	N 65 E, Sub Vert	NO FAULT KNOWN AS SUCH			BAD GROUND Slickensides		
II	N 25-60 W, 70 to 90	CENTER Tiger	N 65° W, Vert N 50° W, Vert or 70° NE	YES YES	NO NO	NO NO	Very little Very little	YES, along the fault YES, along the fault (surface)	YES, irregular ?
		RIVER "C"	N 35 W, 80 to 90	YES	NO	NO	A few inches Slickensides	NO	Unevenly not parallel the
		Yellow Cliff	N 30 W, Vert	YES Cp >> Po	NO	NO	NO	?	YES
III	N 0 to 20 E, vert	BEND	N. S. Vert. to 65° W	NO	NO	YES	YES slickensides water	NO	NO
		River A	N. S. Vert	Not in the fault	NO	YES	Very little	NO	NO
		Forebay	N 0-10 W, Vert	NO	NO	NO	Very little	NO	NO
		Stewart	N 5° W, 75 NE	NO	NO	YES	YES	NO	NO
		Cave	N 0-10° W, Sub Vert	NO	YES	Irregular	YES. 1" to 2"	NO	?
IV	?, Flat	Flat Faults (Lower B, 4th)	Strike Unknown Dip = 0°	NO	NO	YES + Argillite	YES	?	?
SAN JUAN and LEECH RIVER FAULTS (E-W)		NO EQUIVALENT RECOGNIZED							

PART TWO

METALLOGENY

ORE CONTROLS

Four ore-controlling geological features have been described in last year's report (Pages 30 and 31). With the new data gathered and compiled this year, some of these ore controls have been investigated more accurately.

I - CONTACT BASALT - GABBRO (Regional)

It is a fact that several mineralized zones (both uneconomical and economical) are located on or close to the intrusive Gabbro contact zones with the Metchosin Volcanics on the Sooke Map Area.

Nevertheless, there are also numerous mineralized zones located in the Volcanics far away from the intrusions, like the ones currently investigated in the Diversion Reservoir Area (Bear Creek - Alligator Creek), and other zones located within the intrusive itself (Stewart Zones at Jordan River, Cook, Merryth and Huestis Zones at East Sooke). Also, long stretches of contact areas have been investigated in detail without showing any mineralization.

Therefore, the extent to which the gabbro-basalt contact can be considered as an ore control is not established.

More generally speaking, it is not even proven that these contact zones can be used as a guide for regional prospection.

Closer analysis and discussion of this point is given farther in "Hornblendite Zones".

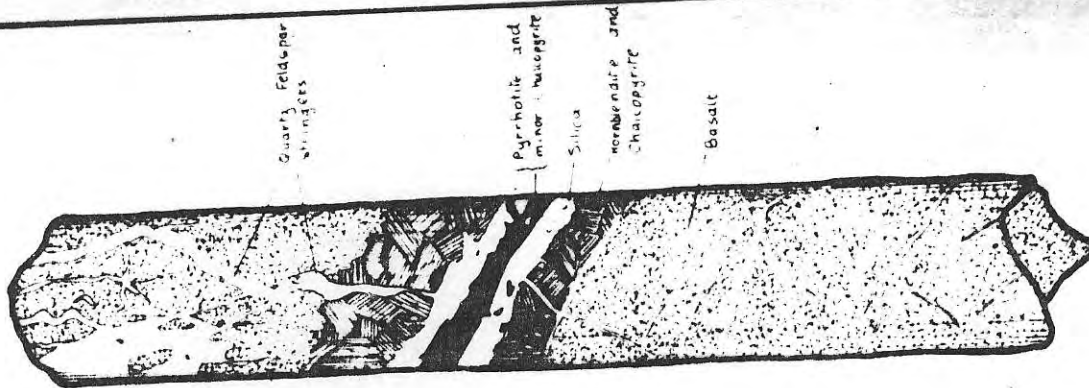
II - HORNBLENDITE ZONES

The presence of Hornblendite, in veins or large zones, had been known for a long time; however, the discontinuity of these zones and their gradational transition to the unaltered gabbro make their identification difficult, especially in the underground workings. These hornblendite zones were therefore not differentiated from the gabbro in the mapping of the mine.

In the last months of this year, a new study of the bibliography has been carried out and a start of mapping this hornblendite has been made, showing the importance of these zones as ore controls.

As a rule, the mineralization of those parts of the Cave Zone which lie within the Gabbro are always associated to hornblendite zones. With very few exceptions, the sulphides are always enveloped by the hornblendite; this of course does not mean that the hornblendite is always mineralized to an economical extent.

Nevertheless, it is established that the hornblendite



Quartz Feldspar
minerals

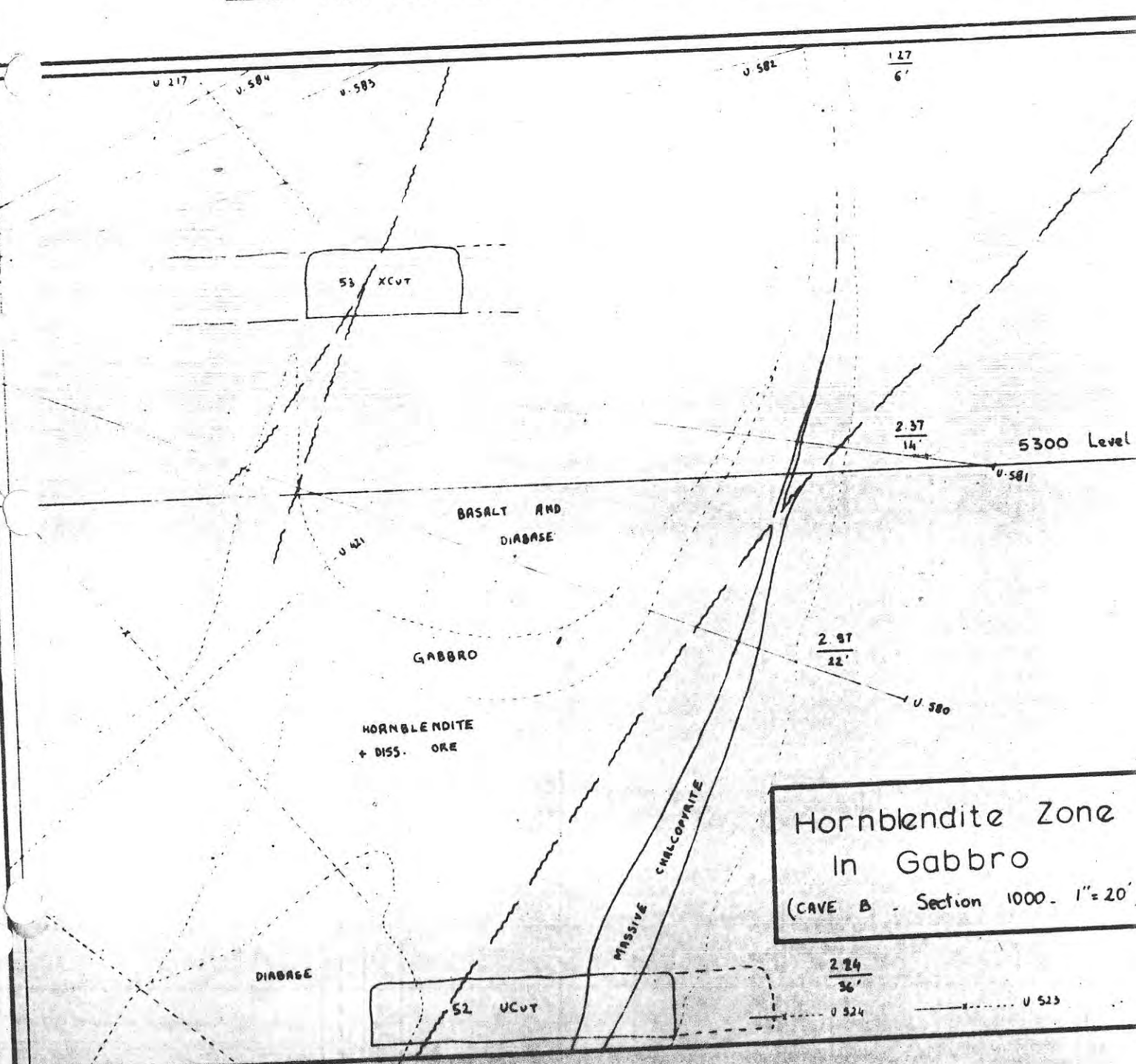
Pyroxene and
minor minerals

Silica

Hornblende and
Chalcopyrite

Basalt

1 cm Hornblende Vein in Basalt (BDH U. 596 - 75')



zones were favourable to ore deposition and they should be used as an exploration guide.

A misleading observation resulted in the idea that the contact gabbro-basalt could be used as an ore control; in fact, in the Cave Zone area the main Hornblendite Zone containing the Cave B Ore Zone is following this contact on a large distance, giving the wrong aspect of a control by the intrusive contact.

III - DIABASE INCLUSIONS

The third geological feature described as an ore control in the 1971 Report was "Fine Grained Hornblendite Inclusions in Gabbro". In fact, these inclusions are diabasic and they have nothing in common with the Hornblendite. These diabase inclusions are always barren and can introduce a heavy internal dilution in the ore zones (Cave B from 52 to 53 on Section 1000, for example).

These inclusions are believed to be pieces of volcanics brought up with the gabbro and metamorphosed. They probably resisted the invasion of the mineralizing solutions.

IV - BASIC DYKES

It has been confirmed that these dykes are barren and are post-ore. (See 1971 Report)

MINERALOGICAL SUCCESSION

The study of numerous ore specimens, underground faces and core samples indicate the following order of crystallization for the different minerals found in the Sunro Orebodies:

↓
HORNBLLENDE
MAGNETITE
PYRRHOTITE

CHALCOPYRITE

QUARTZ - FELDSPARS
PYRITE
↓ NATIVE COPPER

Pyrite occurs either in well shaped cubes (1 to 5 mm) in the faults gouge (Type III) or in thin coatings on slips and joints.

The quartz - Feldspars stringers cut the sulphides stringers (but not the large veins) and some grains are sometimes found in the massive sulphides.

Chalcopyrite and Pyrrhotite are always associated with the Pyrrhotite always cut by the Chalcopyrite.

Both Pyrrhotite and Chalcopyrite stringers cut the large secondary Hornblende Crystals. A few grains of Magnetite are sometimes found in the Hornblendite seams.

MINEGEOLOGICAL ZONING

A - MINE

A - I - GRADE DISTRIBUTION IN THE CAVE ZONE

A start of structural analysis on longitudinal sections has been made for the Cave A and B Zones.

The Cave B is not continuous from 52 to 57, but pinches out around 54 elevation, with a corresponding drop in the copper grade as we go up. However, it seems that mineralized faults and erratic high grade stringers issued from the top part of the Cave B merge into the middle part of the Cave A, resulting in a strong oreshoot called AW2, outlined on 55 and 57 levels.

The Cave A is fairly continuous between 51 and 57, but with several low grade zones.

A good approach for the study of grade distribution is to calculate the average width and grade of the ore zone on each level:

57.....	23.5' @ 1.26
55.....	44' @ 1.33
54.....	Not Enough Data
53.....	Not Enough Data
52.....	45' @ 1.22
51.....	28.5' @ .84

It must be recognized in considering these figures that the data is very incomplete, since the south end of the ore zone has not been reached in any of the mining levels.

However, it seems that the copper grade drops as the depth increases. This conclusion is confirmed so far by the few holes drilled in the Cave Zone below 51 level. (Only three holes, giving respectively 8' @ .80, 10' @ .90 and 39' @ .20 for the deepest one)

On the other hand, one must consider another possibility: the orebody might have a shoot of high grade ore raking approximately 45° to the Southeast. In this case, the study of grade distribution carried out on a vertical plane would be erroneous. Unfortunately, there is not enough information at the present time to confirm this hypothesis.

A - II - GRADE DISTRIBUTION IN THE RIVER ZONE

The main shoot of the River Orebody, the B North, at the junction of the A, C and Cross Fault structures, showed the following

grade distribution before it was mined out:

55 and above:	33.7' @ 1.94	
54	: 48.6' @ 1.50	
53	: 49.8' @ 1.94	
52	: 57.25' @ 1.27	
51	: 39.7' @ 1.10	← 50 : 48.8' @ 2.06
49	: 30' @ 2.15	
48	: 12.6' @ 1.77	

In the case of this orebody, the copper grades distribution does not show obviously a drop with increasing depth, although one will notice that the grades drop from 54 to 51 and then again from 49 to 48.

However, the geological logs of all the diamond drill holes drilled below 49 level indicate an abnormally high amount of Pyrrhotite for the River Zone. Correlatively, some of the samples were assayed for nickel and show relatively high values, up to .372 in U-326 on 4770 elevation.

B - SUNBO PROPERTY

Let us consider the longsection along the main adit (1" - 400'), and the Yellow Cliff, Tiger, Stewart, Cave and River Zones. The grades distribution is as follows:

River Zone.....	1.58
Cave Zone.....	1.26
Stewart Zone.....	.70
Tiger Zone.....	1.38
Yellow Cliff Zone.....	2.62

The Stewart Zone is within the gabbro; the Cave and Tiger Zones are on or close to the contacts; the River and Yellow Cliff Zones are away from the intrusive in the basalts.

It is apparent that there is a relation between the copper grade of the ore zones and their distance to the Central Gabbro Intrusive.

Let us now consider the mineralogy of these zones. The minerals are classified by order of importance, the first one being the most important:

RIVER	Chalcopyrite - Quartz and Feldspars - Pyrrhotite - Very little Hornblende - Very little Pyrite
CAVE	Chalcopyrite - Pyrrhotite - Quartz and Feldspars - Hornblende very abundant - Little Pyrite
STEWART	Hornblende - Pyrrhotite - Chalcopyrite and Pyrite
TIGER	Chalcopyrite - Pyrrhotite - Pyrite - Quartz and

Feldspar - Hornblende stringers and crystals.

YELLOW CLIFF Chalcopyrite (Chalcosite reported but not observed)
Pyrrhotite - Quartz and Feldspar

To summarize this data, the zones in the gabbro and close to the contact have much more Pyrrhotite and Hornblende; on the other hand, Chalcopyrite and Silica are typical of the zones away from the intrusive.

C - SOOKE MAP AREA

Several mineralized zones are known on the area lying south of the Leech River Fault. It seems that they can be classified in three types:

- The Sunro type where Chalcopyrite and Pyrrhotite, Hornblende and Silica Stringers, are characteristic.
- The Alligator Creek type, where Quartz - Feldspars stringers, Chalcopyrite and Pyrite are predominant.
- The Iron Mountain type, which contains chiefly Pyrrhotite, Magnetite and Hornblende.

The geographical distribution of the known showings is very interesting:

On East Sooke peninsula, which is considered by many geologists as the core of the gabbroid intrusion, are found mostly Iron Mountain Type deposits, with a few low-grade Sooke Type orebodies.

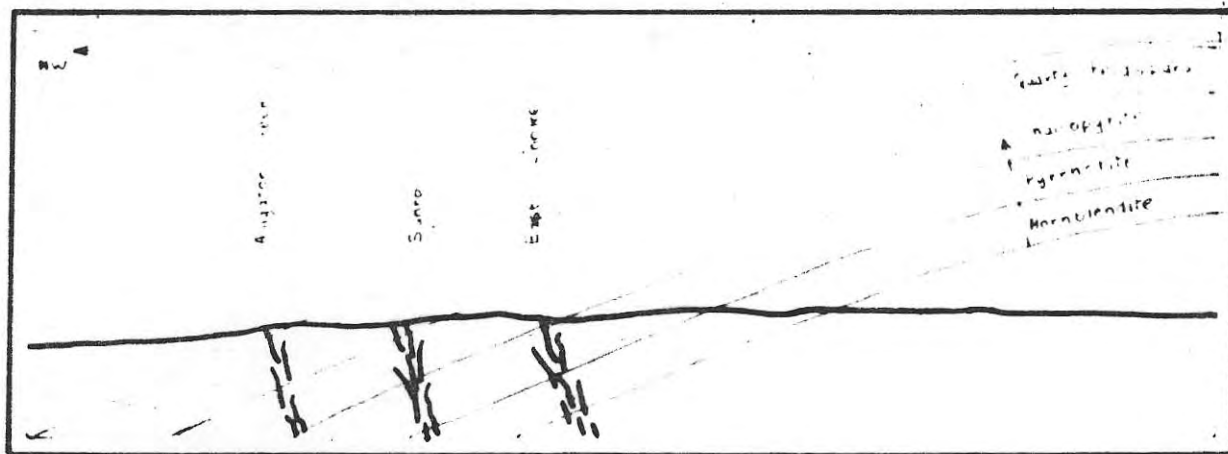
On the Sunro Property are found the Sunro Type orezones, with a few Alligator Creek Type showings (Turnbull Zone - Hornet Zone) and one deposit (Stewart Zone) which resembles the Iron Mountain Type.

In the Diversion Reservoir Area are found the Alligator Type showings.

Here again, it seems that we have a zoning which can be summarized by:

Hornblende —————> Pyrrhotite —————> Chalcopyrite —————>
Silica —————> (~~Magnetite~~) Pyrite

This geographical distribution can be illustrated by the following diagram:



SCHEMATIC DIAGRAM SHOWING THE MINERALOGICAL ZONING
ON SOOKE MAP-AREA

D - SUMMARY AND CONCLUSIONS ABOUT THE ZONING

To sum up the preceding data, we can write:

The zoning Hornblende & Magnetite - Pyrrhotite - Chalcopyrite - Quartz and Feldspars is valid for:

- The vertical axis of the ore zones, from depth to surface
- The distribution of the ore zones across the gabbro intrusive, from centre to edges
- The distribution of the ore zones along the gabbro intrusive, from core to tail end

In other words, we have the same zoning on the three axis X, Y, Z of a reference system in space:

SEE FIGURE NEXT PAGE

3 - The lower extension of the Cave Zone is most likely very poor in copper.

4 - The River Zone could extend at depth by another Cave Zone.

5 - The potential of the Tiger - Yellow Cliff - Caulfield system is excellent, on strike and at depth.

6 - The possibility of an unknown orebody past the River Zone should be studied; this zone being deeply seated would certainly not outcrop.

It must be kept in mind that the potential of the ore-zones at depth, as mentioned in 2, 4, 5 and 6, depends only on the continuity of the shear zones in which the hornblendite zones and mineralization are contained. This goes back to the Tectonic Study started in 1971.

As far as No. 6 conclusion goes, it is very interesting to notice the presence of a fairly well defined shear zone exposed in a creek approximately 600 feet northeast of the River Zone. This zone, located in the basalts, shows a heavy gossan with a few Pyrite occurrences and a loose network of Quartz and Feldspars stringers; this could be considered as the upper termination of a deeply seated orebody, although no copper mineralization can be found on the surficial showings. Assuming a dip similar to the one of the River Zone, and keeping the same mineralized height above the projected gabbro-basalt contact than for the River Zone, the apex of the copper bearing mineralization would be located approximately at 1,000 feet below the surface: i.e. on 4600 elevation. The ore zone would extend down to 3000 elevation (2600 feet below surface).

ORE RESERVES

I - METHOD OF CALCULATION

The techniques used to compile and calculate the ore reserves in 1972 are the same as those used previously.

Definitions of the terms Proven, Probable and Possible Ores have been explained in "Report on Ore Reserves Estimation" by Nicolet, Guelpa and Meusy - February, 1971.

Calculation has been made using cross-sections for the Cave, River A and Hanna Zones. For the River B, River C and New Zones, longitudinal sections have been used.

II - CUT-OFF GRADES

The cut-off grades have been calculated as follows:

BASIC DATA

Mining and Milling at 1500 tons/day
Mill Recovery - 93%
Concentrate Grade - 27%
Smelter Contract as per Memo dated
October 19/1971 - Modified January 13/1972

CHART ONE - CONCENTRATE VALUE

LME Copper Price	\$/Dry Metric Ton	\$/Short Ton
55¢ /Lb.	228.19	207.01
50¢ /Lb.	199.76	181.22
45¢ /Lb.	171.33	155.42
40¢ /Lb.	142.88	129.62

CHART TWO - NET SMELTER RETURN PER ORIGINAL POUND OF COPPER AT 93% RECOVERY

LME Copper Price	NSR /Lb.
60	41¢
55	36¢
50	31¢
45	26¢
40	22¢

65 ¢ 46 ¢

CHART THREE - PRODUCTION RECEIPTS
(NET SMELTER RETURN AT MINE PER TON MILLED)

Mill Heads	40¢	45¢	50¢	55¢	60¢	65¢
1.50	6.60	7.80	9.30	10.88		
1.45	6.38	7.54	8.99	10.41		
1.40	6.16	7.28	8.68	10.01		
1.35	5.94	7.02	8.37	9.72		
1.30	5.72	6.76	8.06	9.36		
1.25	5.50	6.50	7.75	9.00		
1.20	5.28	6.24	7.44	8.64		
1.15	5.06	5.98	7.13	8.28		
1.10	4.84	5.72	6.82	7.92		
1.05	4.62	5.46	6.51	7.56		
1.00	4.40	5.20	6.20	7.20		
.95	4.18	4.95	5.89	6.84		
.90	3.96	4.68	5.58	6.48		

CHART FOUR - COSTS ANALYSIS (Figures Communicated)

	LONGHOLE STOPES	SHRINKAGE STOPES
MINING	.85	1.45
MILLING	1.50	1.50
OVERHEAD	1.00	1.00
ROYALTY	.35	.35
COST/TON	3.70	4.30

Assuming a profit of \$2.00 per ton, the cut-off grades are:

LME Copper	55¢	50¢	45¢	40¢
L.H.	.80	.92	1.09	1.29
SHRINK	.87	1.02	1.21	1.44

Taking an average LME Copper Price of 47¢, the cut-off grades utilized for calculations are 1.02 for L.H. and 1.13 for Shrink.

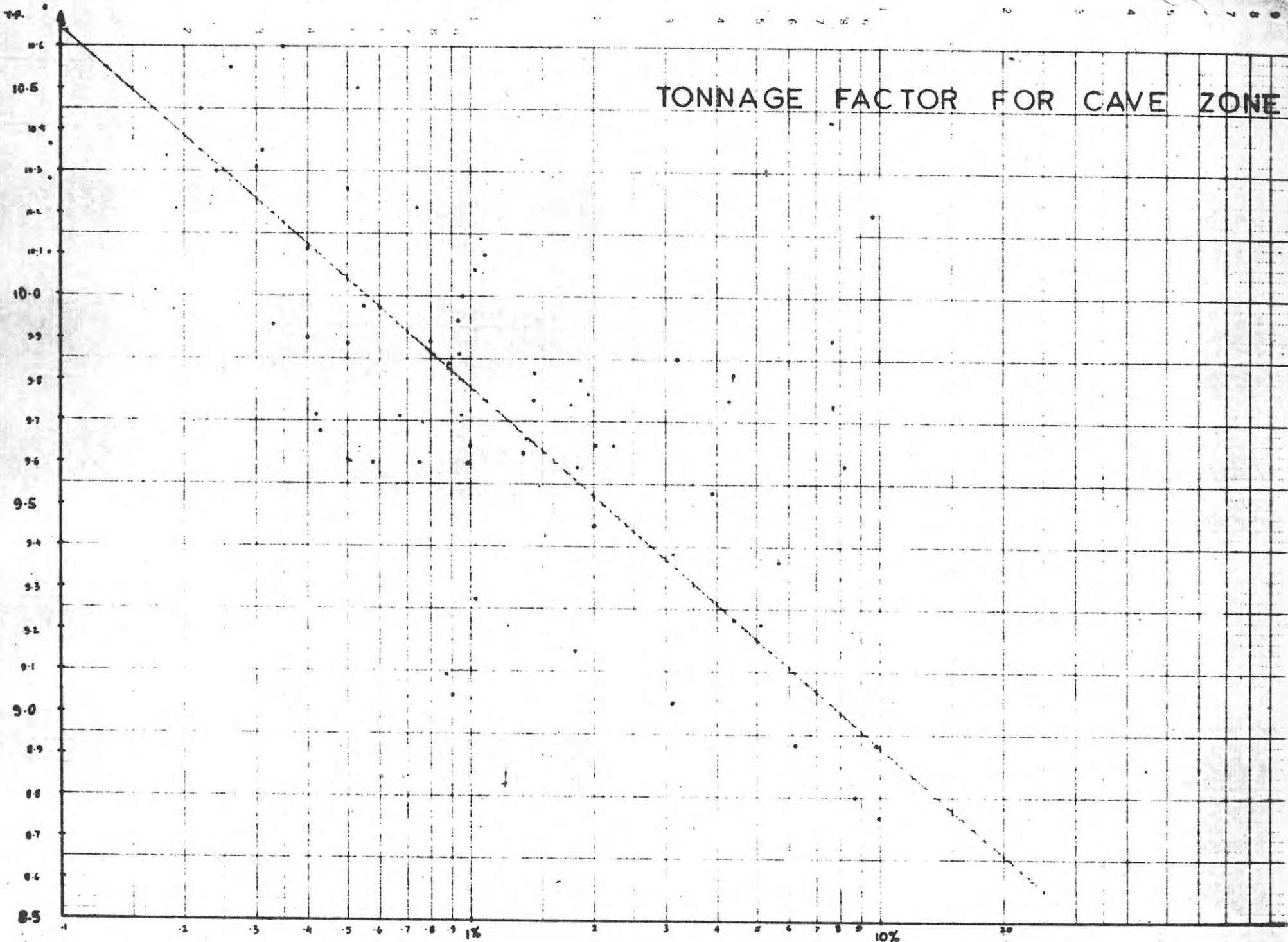
III - MINIMUM MINING WIDTH

The following assumptions have been accepted: for shrinking, minimum width is 4 feet; for longholing, the minimum mining width is 12 feet. All grades given by an assay shorter than the minimum mining width have been purposely diluted to reach this minimum width.

IV - TONNAGE FACTOR

A new series of samples has been taken in the Cave Zone to build the curve Tonnage Factor = $f(\text{Copper Grade})$. The readings are still erratic. Another graphic has been drawn with Tonnage Factor = $f(\text{Ratio } \frac{\text{Fe}}{\text{Cu}})$.

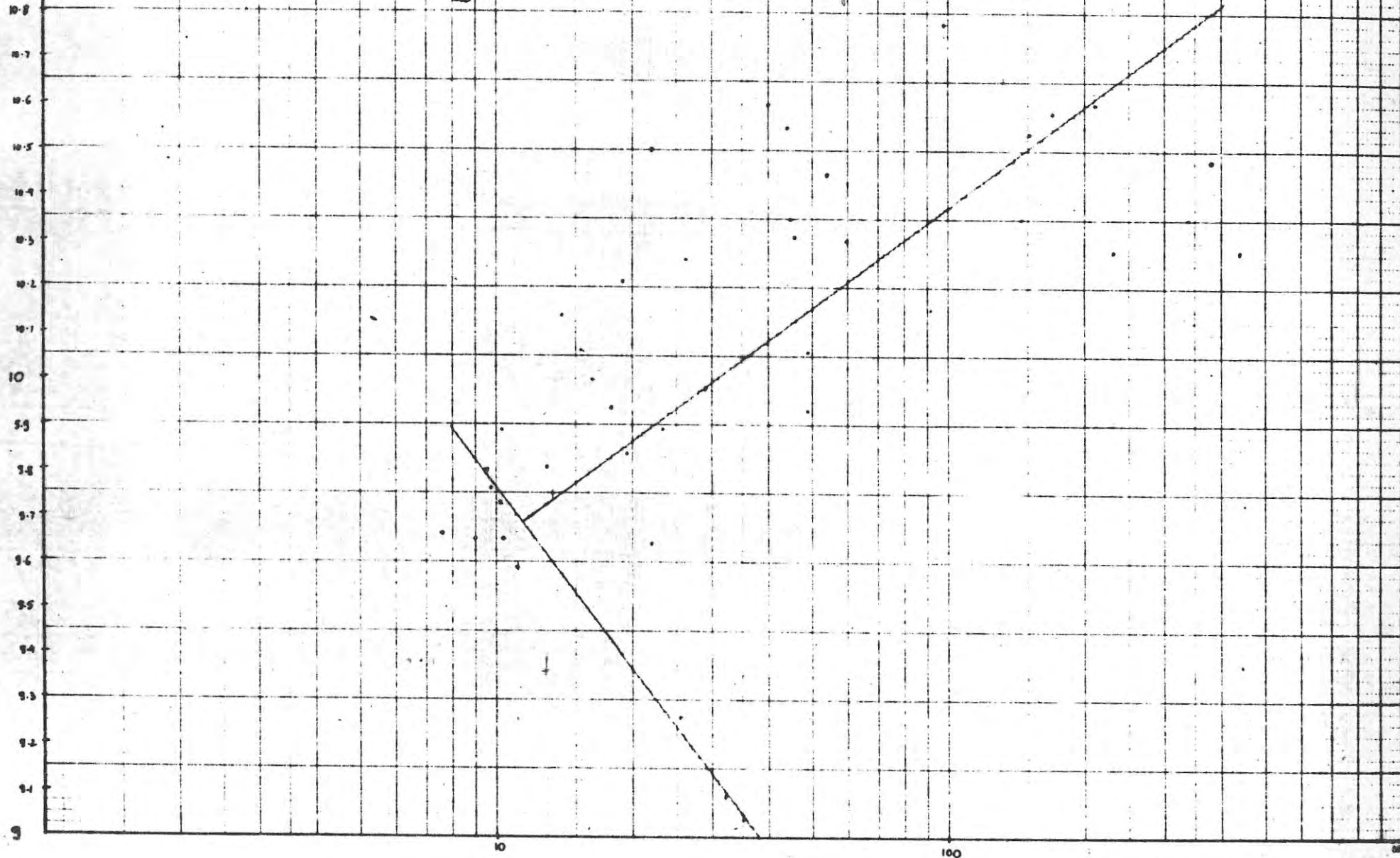
This graph seems to indicate two lines at right angle - for the low grade values, the density goes up when the copper grade drops, and for the medium and high grade readings, the density increases when the copper grade goes up. An average tonnage factor of 10 has been used for the Cave Zone, and a variable factor has been used for the other Zones, according to the River Zone Tonnage Factor Curve. (See 1971 Report)



T.F.

TONNAGE FACTOR FOR CAVE ZONE

$$TF = P \left(\frac{R}{P} \right)$$



ORE RESERVES CRITICAL STUDY

This chapter is intended to give an estimate of the ore reserves readily available in the mine.

I - RIVER ZONE

I - A - RIVER A

A total figure of 160,213 Tons @ 1.41 is shown on the Ore Reserve Chart. In considering this tonnage, one should keep in mind the following facts:

- A1 and A3 lenses are only "horsetails" of either the B or the A Zones. They show a very erratic mineralization, with extremely punctual high values. On the average, these high assays are under four feet wide. These two lenses must therefore be considered as non-minable.

- Due to the River B North caved stope pillar, the A ore lying to the northwest of Section 1600 is not recoverable at the present time.

The available reserves in the A2 lens are therefore:

SECTION 1500	3,000 T @ 1.23
SECTION 1550	20,666 T @ 1.56
SECTION 1600	12,662 T @ 1.18
TOTAL PROVEN	36,328 T @ 1.40
+ PROBABLE	12,262 T @ 1.07
TOTAL MINABLE ORE	48,590 T @ 1.32

This figure actually represents the whole River A Zone which can be mined today: i.e., 30% of the total reserves.

I - B - RIVER B

Due to the condition of the B North Stope, the ore tied up in the sill and around the Service Raise is not recoverable at the present time.

The B Southeast lens is situated between the B South Shrink Stope and the empty C longholed stope. It is known as only 100 feet long. It does not seem possible to establish a stope on this lens.

Finally, it appears that most of the ore in the lower B is also tied up in pillars below the B North and around the small stopes of the shaft area.

SUNRO MINE

ORE RESERVES

DATE JAN 5 1973

ZONE & BLOCK		HEIGHT	LENGTH	BROKEN			PROVEN			PROBABLE			TOTAL		
Name	Feet	Feet	TONS	% Cu	T x G	TONS	% Cu	T x G	TONS	% Cu	T x G	TONS	% Cu	T x G	
CAVE A	570	800	5,500	1.01	5,555	412,975	1.16	481,853	286,935	1.24	355,917	705,410	1.19	843,325	
CAVE B	400	500	4,100	1.28	5,248	121,755	1.55	189,128	38,550	1.42	54,873	164,405	1.52	249,249	
TOTAL CAVE ZONE			9,600	1.12	10,803	534,730	1.25	670,981	325,485	1.26	410,790	869,815	1.26	1,092,574	
RIVER A 1	470	100	Nil	-	-	10,220	2.40	24,528	Nil	-	-	10,220	2.40	24,528	
RIVER A 2	400	200	Nil	-	-	120,998	1.32	159,997	18,762	1.37	25,766	139,760	1.33	185,763	
RIVER A 3	270	75	Nil	-	-	10,233	1.56	15,963	Nil	-	-	10,233	1.56	15,963	
TOTAL RIVER A ZONE			Nil	-	-	141,451	1.42	200,488	18,762	1.37	25,766	160,213	1.41	226,254	
B SERVICE RSE PILLAR	470	50	Nil	-	-	91,900	1.89	173,691	Nil	-	-	91,900	1.89	173,691	
B SILL PILLAR	100	190	Nil	-	-	30,100	2.25	67,725	Nil	-	-	30,100	2.25	67,725	
LOWER B	350	200	Nil	-	-	200,000	1.84	368,000	Nil	-	-	200,000	1.84	368,000	
B SOUTH	200	180	Nil	-	-	18,653	2.34	43,726	18,192	2.35	42,835	36,845	2.35	86,561	
B SOUTH-EAST	470	100	Nil	-	-	24,191	1.60	38,705	Nil	-	-	24,191	1.60	38,705	
TOTAL RIVER B ZONE			Nil	-	-	364,844	1.90	691,847	18,192	2.35	42,835	383,036	1.92	734,682	
UPPER C	700	570	Nil	-	-	53,746	1.81	97,382	46,982	1.52	71,635	100,728	1.68	169,017	
LOWER C	200	100	Nil	-	-	Nil	-	-	32,500	1.76	57,200	32,500	1.76	57,200	
TOTAL RIVER C ZONE			Nil	-	-	53,746	1.81	97,382	79,482	1.62	128,835	133,228	1.70	226,217	
TOTAL RIVER ZONE			Nil	-	-	560,041	1.77	989,717	116,436	1.69	197,436	676,477	1.75	1,187,153	
TOTAL CAVE & RIVER			9,600	1.12	10,803	1,094,771	1.52	1,660,698	441,921	1.38	608,226	1,546,292	1.47	2,279,727	
NEW ZONE	150	330	Nil	-	-	Nil	-	-	25,844	1.67	43,320	25,844	1.67	43,320	
HANNA ZONE	200	200	Nil	-	-	Nil	-	-	94,160	1.12	106,132	94,160	1.12	106,132	
SUB-TOTALS			9,600	1.12	10,803	1,094,771	1.52	1,660,698	561,925	1.35	757,678	1,666,296	1.46	2,429,179	

TOTAL RESERVES INCLUDING 20% DILUTION @ .15 - 1,999,554 TONS @ 1.24

Calculated by : G. Meusy

The B ore minable above 52 level is therefore:

30,845 Tons @ 2.35

This is 10% of the total reserves.

Assuming that the ore of the Lower F can be recovered, the total available River B reserves are:

236,845 Tons @ 1.92

This figure represents 62% of the total reserves.

I - C - RIVER C

All the ore figured out for the upper C Zone is minable at the present time. There is no apparent restriction for the lower C either. The available tonnage is therefore 100% of the total reserves.

II - CAVE ZONE

II - A - CAVE A

The ore lying below 5225 level amounts to 169,895 Tons @ 1.25. This ore will not be recoverable until the upper Cave is completely mined out and a new mining system is established below the 51 horizon.

Available reserves thus become:

535,515 Tons @ 1.18 (Proven and Probable)

This represents 76% of the total reserves.

II - B - CAVE B

The Cave B ore included in the reserves but lying below the haulage level 5225 is 71,210 Tons @ 1.39. This ore is subject to the same restrictions as for the Cave A.

Available reserves in the Cave B must therefore be read:

93,195 Tons @ 1.61

This is 57% of the total reserves.

III - SUMMARY

Since there is no restriction for the recently developed New Zone tonnage, the total available reserves in the mine are:

CAVE A	535,515 Tons @ 1.18
CAVE B	93,195 Tons @ 1.61
RIVER A	48,590 Tons @ 1.32
RIVER B	36,845 Tons @ 2.35
RIVER C	33,228 Tons @ 1.70
NEW	25,844 Tons @ 1.67

TOTAL UNDILUTED	873,217 Tons @ 1.38
-----------------	---------------------

TOTAL WITH 20% DILUTION @ .15	1,047,860 Tons @ 1.17
----------------------------------	-----------------------

Tonnagewise, this represents only 52% of the total diluted reserves.

A FEW FIGURES ON POSSIBLE ORE

No possible ore has been taken in account in any of the ore reserves statements. Without going too far, one can estimate that the possible ore in the Cave and River Zones is closer to the Probable type than in the other orebodies.

Without taking in consideration the possibility of finding ore below the 51 level (see chapter on Exploration Proposals) and below the shaft area, the following figures should be retained for possible ore:

RIVER A	- No possible ore above 51
RIVER B	- 575 Tons @ 1.22
RIVER C	- 10,207 Tons @ 1.22
CAVE A	- 210,000 Tons @ 1.05 (54 to 57) 269,000 Tons @ 1.10 (above 57)
CAVE B	- 10,000 Tons @ 1.25 (above 53)

TOTAL POSSIBLE ORE IN CAVE AND RIVER BETWEEN PRESENT MINING LEVELS	- 499,782 Tons @ 1.08
--	-----------------------

ORE RESERVES STATISTICAL STUDY

This chapter is intended to gather the data on ore reserves in order to build the Ilasky Curve of the Sunco Deposit.

I - TOTAL ORE MINED OUT TO DATE

According to the available records, the ore mined out in the River Zone is:

B NORTH -	336,997 Tons @ 1.70
RIVER C -	342,703 Tons @ 1.30
LOWER B -	159,284 Tons @ 1.32

In 1972 ore was mined out from the Cave Zone:

CAVE A -	41,608 Tons @ 1.06
CAVE B -	51,649 Tons @ 1.33

II - TOTAL RESERVES

Adding the broken, proven and probable reserves to the mined out ore, the total figures become:

CAVE A -	747,018 Tons @ 1.19
CAVE B -	216,054 Tons @ 1.47
RIVER A -	160,213 Tons @ 1.41
B NORTH &	
LOWER B -	696,281 Tons @ 1.65
RIVER C -	475,931 Tons @ 1.41

III - CUMULATED RESERVES

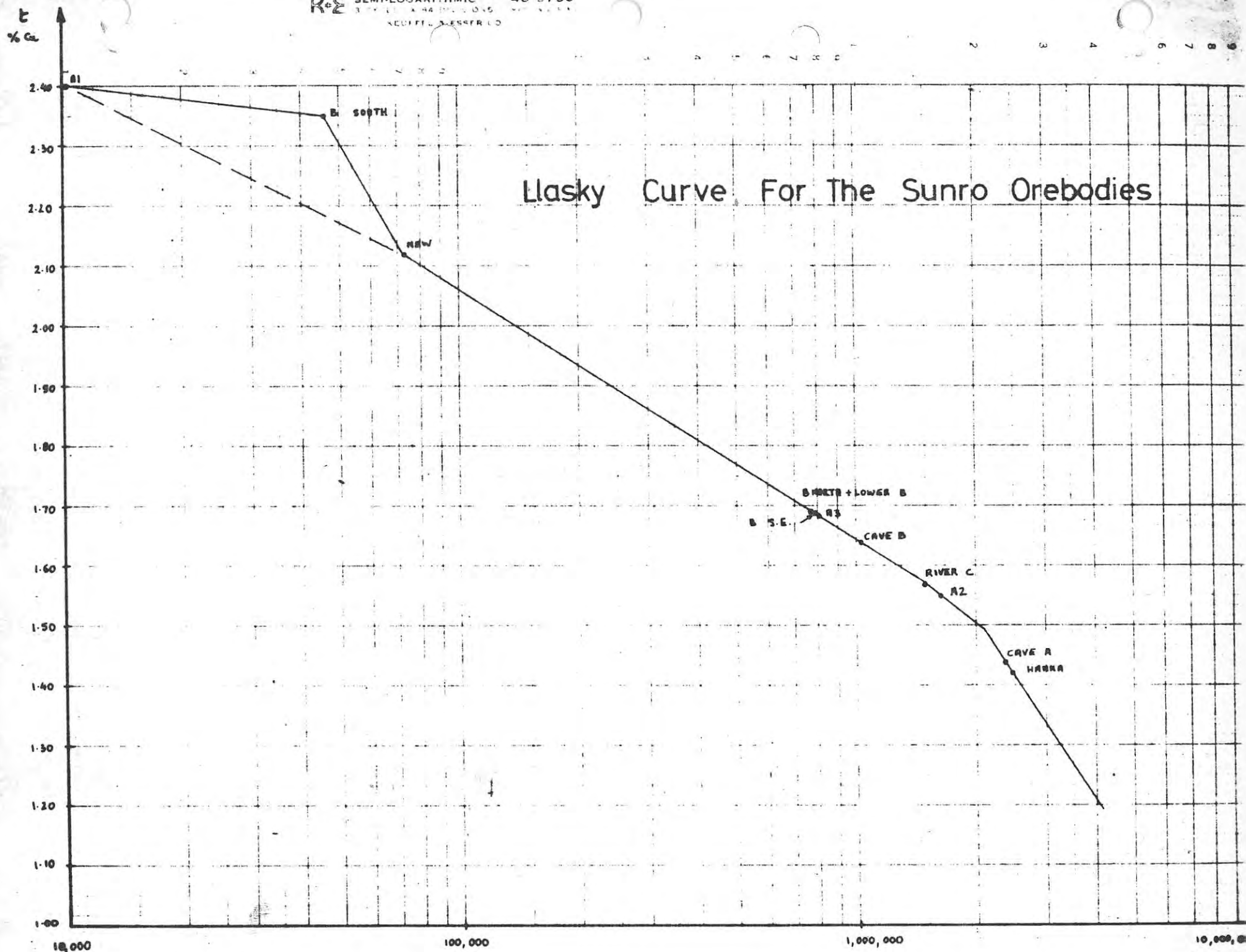
To build this chart, the ore reserves figures have been classified by decreasing grade and cumulated with weighted tonnages and grades.

ZONE	TONS	GRADE	T x Gr	C. TONS	AVG. GR.	T x Gr
RIVER A1	10,220	2.40	24,528	10,220	2.40	24,528
B SOUTH	36,845	2.35	86,561	47,065	2.36	111,089
NEW ZONE	25,844	1.67	43,320	72,909	2.12	154,409
B NORTH & LOWER B	696,281	1.65	1,148,864	769,190	1.69	1,303,273
B SOUTHEAST	24,191	1.60	38,705	793,381	1.69	1,341,978
RIVER A3	10,233	1.56	15,963	803,614	1.69	1,357,941
CAVE B	216,054	1.47	317,599	1,019,668	1.64	1,675,540
RIVER C	475,931	1.41	671,063	1,495,599	1.57	2,346,603
RIVER A2	139,760	1.33	185,763	1,635,359	1.55	2,532,366
CAVE A	747,018	1.19	888,951	2,382,377	1.44	3,421,317
HANNA	94,160	1.12	106,132	2,476,537	1.42	3,527,449

IV - LLASKY CURVE

The curve build according to the above data is presented next page.

The lower part of the curve shows a change in the slope, which would indicate that the Cave A and Hanna Zones are a different type of orezone than the others. Probably many more figures on tonnages and grades for the other orebodies are required.



PART THREE

EXPLORATION

SURFACE STANDING

At the time when this report was written, the mining property was composed of the following claims:

- 52 SUNRO CLAIMS (Full-size and Fractions) owned by Sunro Mines Ltd. - Crown Granted
- 5 Located Claims (Sun Fr 1, Sun 3, Gabbro Fr 2, 4 and 6) Held by Cominco. Expiry Date - March 14th, 1973
- 20 COOK CLAIMS - Located - Recorded on October 15th, 1971 and held in a group with Crown Granted Claims (See further)
- 14 RED CLAIMS - Located and held until March 6th, 1973.

The Crown-Granted Sunro Claims are subject to a number of reservations as shown on the following map.

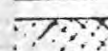
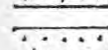
Reservations on base metal rights also apply to all the claims within Lot 4, Section 3 and 7, i.e.: the 14 RED Claims and the 5 Located Claims held by Cominco.

The 20 COOK Claims have been grouped with 6 Crown-Granted Claims (see copy of the Notice to Group) to form the SUNCOOK GROUP. An application for Certificates of Work covering these claims for 10 years has been filed on October 10th, 1972, using \$20,000 spent in the mine area on Sunlock No. 6 Claim.

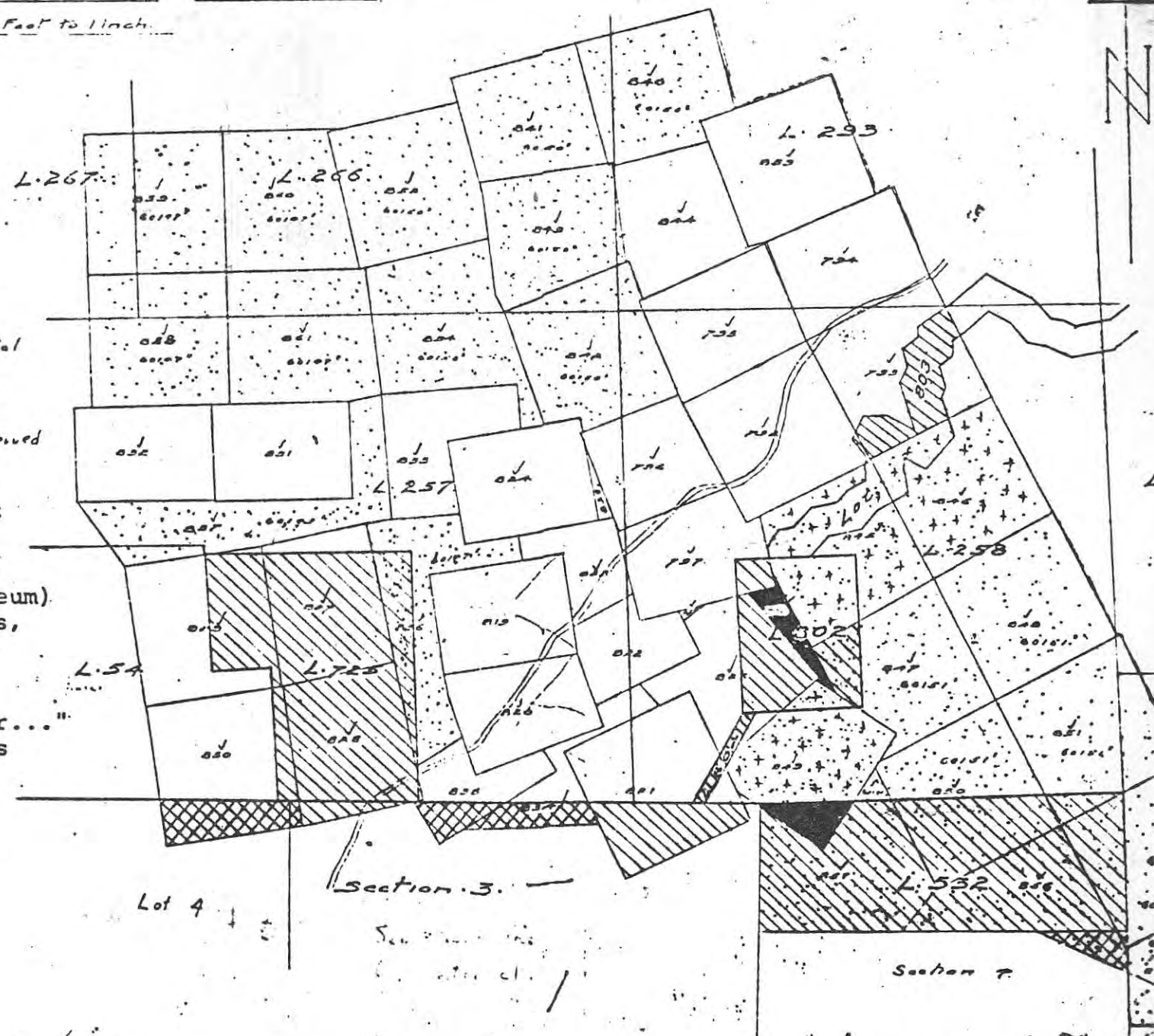
A similar procedure will be used in March, 1973, to group the RED Claims, the 5 Located Claims (Sun and Gabbro) and 4 Crown-Granted Claims. On this Group will be applied work done underground on Gabbro Fraction.

— Mineral Claims on Jordan River —

Scale: 1000 Feet to 1 inch.

-  timber reserved.
-  surface and timber reserved.
-  surface, timber and base metal reserved.
-  mineral rights only granted - no reservations.
-  timber and natural gas reserved.
-  mineral rights only granted, timber reserved.

These Crown Grants include "all minerals precious and base (save coal and petroleum) which may be found in veins, lodes or rock in place and whether such minerals are found separately or in combination with each other..." subject to the reservations to the Crown as indicated.



13 - Raymer has yellow shaded base minerals

SUMMARY OF EXPLORATION DONE IN 1972

I - SURFACE

A small MAG survey had been conducted on the Stewart Zone area in September by J.P. Guelpa.

The results were very inconclusive. A small anomaly with 880 gammas amplitude has been outlined above diamond drill holes 13 and 14. No anomaly has been found past the known showings and the northern extension of the Zone, if any, remains undiscovered.

At the same time, a MAG survey and a Soil Sampling Survey were started on the COOK Claims, No. 8, 10, 12 and 14. All results were negative.

II - UNDERGROUND

Only two areas have been investigated besides the Cave and River Zones:

II - 1 - NEW ZONE

18 diamond drill holes totalling 3,673 feet and 130 feet of exploration drift have been worked out to investigate the New Zone.

Diamond Drilling Results-

HOLE	LENGTH	LEVEL	NEW ZONE
U-493	60'	56	7' @ .92
U-494	72'	56	3' @ 3.54
U-495	70'	56	BLANK
U-496	60'	56	BLANK
U-497	60'	56	2' @ .50
U-521	251	56	10' @ 1.46
U-522	280	56	5' @ .37
U-526	304	55	BLANK
U-527	283	57	BLANK
U-529	300	53	1' @ 7.31
U-532	300	53	3' @ 1.10
U-533	228	53	3' @ 4.26
U-534	253	56	BLANK
U-535	252	56	BLANK
U-536	251	56	3' @ .76
U-537	205	53	BLANK
U-538	195	53	4' @ .72
U-539	249	57	8' @ 9.11

Drifting Results

The 130 feet of drift driven on the New Zone from the downramp on 5570 elevation gave an average grade of 8' @ 1.30.

Conclusion

Much more work is required on this zone between 53 and 55 levels, and below 51. With the exception of a high grade ore shoot outlined on 5670 by U-22 and U-539 and on 5600 by U-521, the New Zone seems to be composed of a series of low-grade lenses without any significant extension. These sulphides lenses are located in a shear zone making an angle of 30° with the River C Zone. Diabase "puffs" are associated to this shear zone.

II - 2 - HANNA ZONE

Four diamond drill holes have been laid out and completed on the Hanna Zone:

HOLE	COORDINATES	ELEVATION	BEARING	DIP	LENGTH	RESULT
U-550	12,581.76 N 12,630.65 E	5530.50	S 72° 50' W	FLAT	465	BLANK
U-551	12,579.58 N 12,631.95 E	5530.39	S 44° 02' W	FLAT	453	BLANK
U-600	12,503.17 N 12,581.00 E	5329.56	S 38° 22' W	FLAT	555	BLANK
U-601	12,503.31 N 12,581.05 E	5328.57	S 38° 40' W	-10°	504	$\frac{1.09}{11'}$

These results allow us to draw two important conclusions:

- There is no relation between the so-called Hanna showing on Surface and the Hanna Zone exposed in the 51 Adit.
- There is no relation between this Hanna Showing on surface and the Cave B Block.

The Hanna Zone appears to be a low-tonnage medium grade zone, with a somewhat erratic grade distribution. The general strike of the Zone is N 120-125 E and its dip is 40-45° NE.

EXPLORATION PROPOSALS FOR 1973

I - INTRODUCTION

In considering any exploration program, one has to define the reasons justifying the program, and the problems it will create.

I - 1) Reasons for Exploration

There are two main requirements:

- 1 - a - New orebodies have to be found, investigated and partially developed before the known orezones are nearing exhaustion.
- 1 - b - In order to maintain production until new mining areas are opened up by 1 - a, all extensions and major off-shoots of the known orebodies have to be outlined.

I - 2) Problems Created by Exploration

In any exploration proposal involving drifting, the main problem encountered is muck disposal. If the muck extracted is ore, it will have to be either trammed to the mill, or stockpiled on surface. If it is waste, it would have to be hauled outside, since the waste storage capacity underground is very limited.

Two other problems are to be solved:

- For drifting or diamond drilling from the present underground network, air and water supplies are critical because of the distances to the compressors.
- For deep diamond drilling, deviation in the order of 1° 30' per 100 feet has to be considered.

In laying out the exploration proposals presented in the following pages, three guidelines dictated by the above requirements and problems have been observed:

- 1) Study two Exploration Programs, one answering to 1-a, labelled "LONG RANGE EXPLORATION" and another one called "SHORT RANGE EXPLORATION", required by 1-b.
- 2) Avoid as much as possible the underground approach for the areas far away from the present mill set-up. Favour proposals involving surface diamond drilling and driving adits.
- 2) Unless absolutely necessary, avoid deep diamond drill holes on relatively small targets.

II - LONG RANGE EXPLORATION

Out of the three proposals in this program, two are based on Geological Theories only.

II - 1 - AREA BELOW RIVER ZONE (MAP EXP II-1)

BASIC DATA

Consequence of the geological theory according to which the River Zone would extend at depth by an orebody similar to the Cave Zone.

As shown on map EXP-II-1, the only way to investigate this area is by diamond drilling. According to the depth of investigation wanted, two programs can be outlined:

II - 1 - A Depth of investigation - 500 feet below the last information in the River Zone (bottom of winze)

This would require two deep holes drilled from the Hanna Zone Cross-cut. The deepest would have to reach 2,000 feet, the other 1,500 ft. In the most favourable case where expectations are confirmed, these two holes could indicate close to one million tons of possible ore. On the other hand, deviation and cost for such deep holes are two major problems. Very little data on deviation in deep holes is available at the mine. (See Technical Part - Diamond Drilling) An average dip deviation of $1^{\circ}30'$ per 100 feet has been used, giving a total of 30° for 2,000 feet. Azimuthal deviation is even harder to estimate, since records indicate figures of 9° for a 1,060 ft. hole to 15° for a 1,100 ft. hole. None of these holes has penetrated the gabbro-basalt contact. In order to reach the target, it is highly probable that wedging would have to be used.

To drill a 2,000 foot hole successfully, a large capacity machine is required, not smaller than a Longyear 38 or BBS 35-AUG. A Longyear 38 would use Diesel or Electric power, whereas a BBS 35-AUG could use compressed air. The size of the hole should be AX or AQ, not smaller.

For such deep drilling, cost is estimated at \$12 per foot, giving a total cost for this proposal of \$42,000.

II - 1 - B Depth of investigation - bottom of the winze

This program would involve four holes in the lower extension of the River Zone (South of the winze) and two holes in the lower Cave Zone.

River 1	-40°	1,100'
River 2	-40°	850'
River 3	-40°	850'
River 4	-40°	1,000'
Cave 1	-50°	1,000'
Cave 2	-45°	1,000'
Total footage		5,800'

12 900

GABRO

JORDAN RIVER

VOLCANICS

5000 (Sea

RIVER 1
1,100'
original Dip
-40°

CAVE 1
original
Dip - 60°

2,000 ft
original
Dip - 40°

Proposal II-1-A

Proposal II-1-B

Section along 51 Adit

JORDAN RIVER

(II-1-A)
2,000 ft

400 ft Xcut

RIVER 1

1,100'

-40°

(II-1-A)

1,500 ft

RIVER 2

850'

-40°

RIVER 3

650'

-40°

CAVE 1

1,000'

-60°

CAVE 2

1,000'

-40°

4,000 39M

EXP II.1

1" = 400'

In the River Zone area, although the bottom of the winze at 4600 elevation is still in ore (observation reported - not seen by the writer), no diamond drill holes information exists below 4800 level; and from 4800 to 5000, information is nonexistent for the southern part of the River Zone.

Structural analysis on the River C indicates a strong shoot raking at 33° Southeast, which would bring the extension of the zone at approximately 1,000 feet southeast of the winze on 4600 elevation. This explains why the four holes on the River Zone have been laid out on the southern extension of the orebody.

The two holes in the lower Cave have also been laid out on the southern part of the orebody for similar reasons.

The deepest hole in this program is River 1 with 1,100 (1,100) feet. Deviation for such a depth would be around 15°, which would not be a serious problem.

To handle this drilling, a BBU-2 equipped with dual 6" Rod Puller is suitable.

The three set-ups are easily accessible and at the most 1,000 feet away from the underground compressors.

Cost is estimated at \$8. per foot, giving a total of \$46,400.

In the most favourable case, this program could indicate 300,000 tons of possible ore in the River Zone and 500,000 tons in the Cave (probably low-grade).

II - 2 - YELLOW CLIFF - CAULFIELD - TIGER SYSTEM (MAP EXP II-2)

BASIC DATA:

Tectonic Trap similar to the River Zone. A proposal has been presented for this area in 1971 Report. ("Exploration Proposal No. 1" - Page 7) This proposal has the disadvantage of starting a drift underground from the 51 Adit, creating much disposal problems.

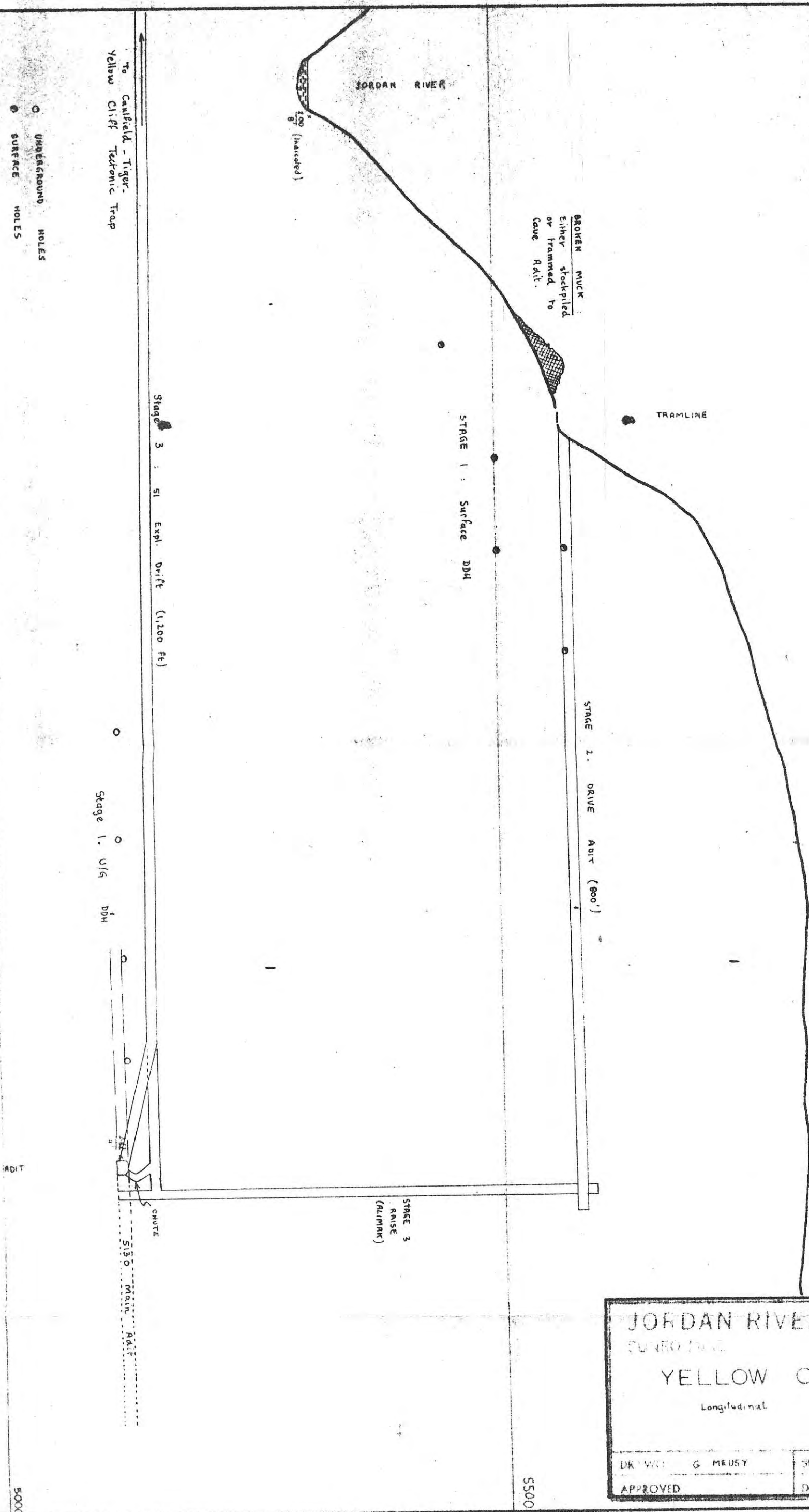
An alternative program could be conducted as follows:

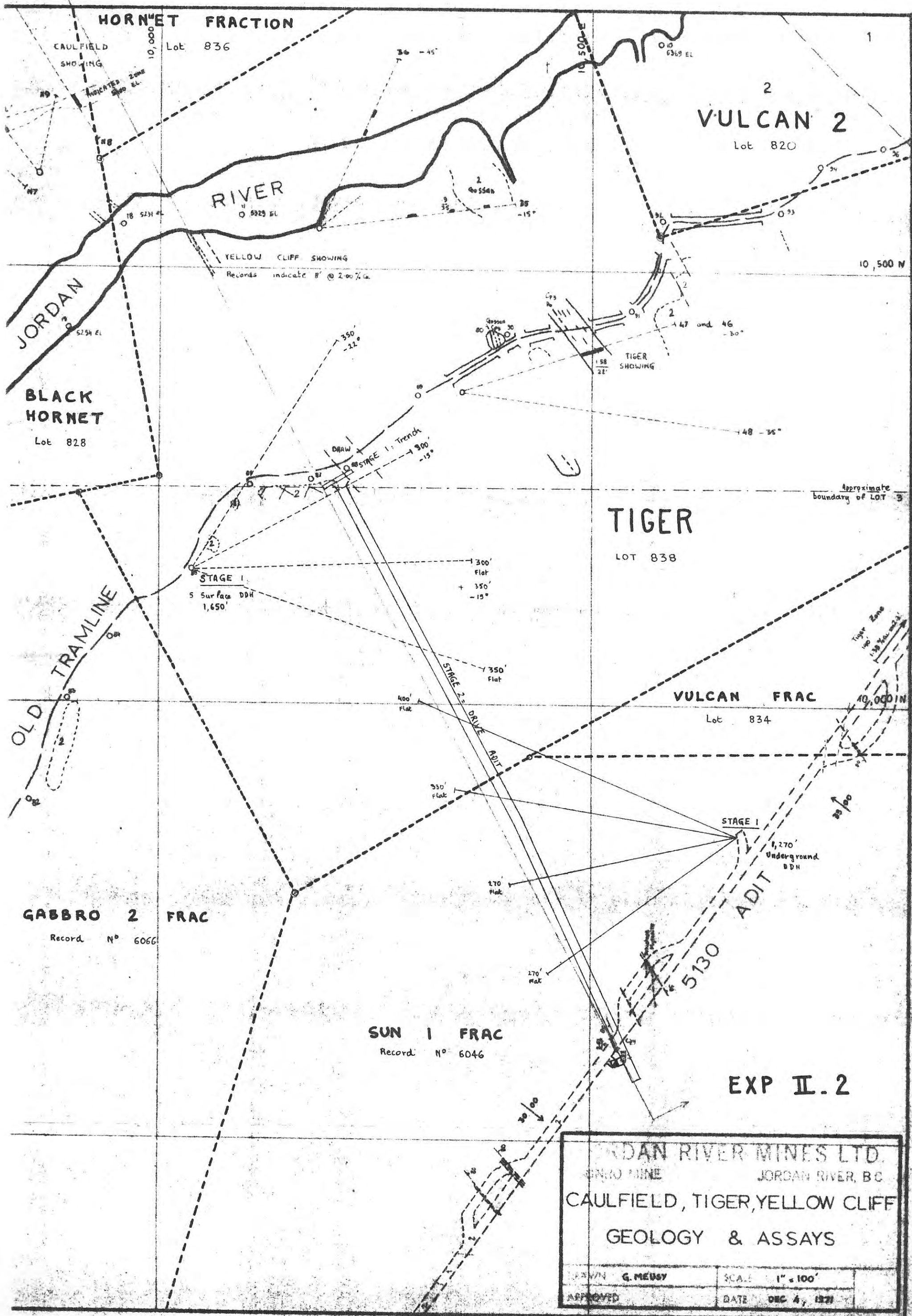
Stage 1: From the 51 Adit, set-up 3400, drill 4 holes on the Yellow Cliff extension to obtain information on strike at 5130 elevation. This job can be handled by a JV with 4½" Rod Puller; a portable compressor equipped with a scrubber would be the easiest way for compressed air supply. Water could be pumped from the ditch and re-circulated.
COST: 1,270 feet @ \$3. - \$3,810
TIME: 3 weeks

From the old surface tramline, Station 85, drill 5 AX holes to gain information on the Yellow Cliff Zone on 55 and 56 levels. This job involves 1,650' of

EXP II.2

JORDAN RIVER MINES LTD.	
JORDAN RIVER, B.C.	
YELLOW CLIFF ZONE	
Longitudinal Section	
DR. WIT: G. MEUSY	SCALE: 1" = 100'
APPROVED	DATE: DEC 19 th , 1972

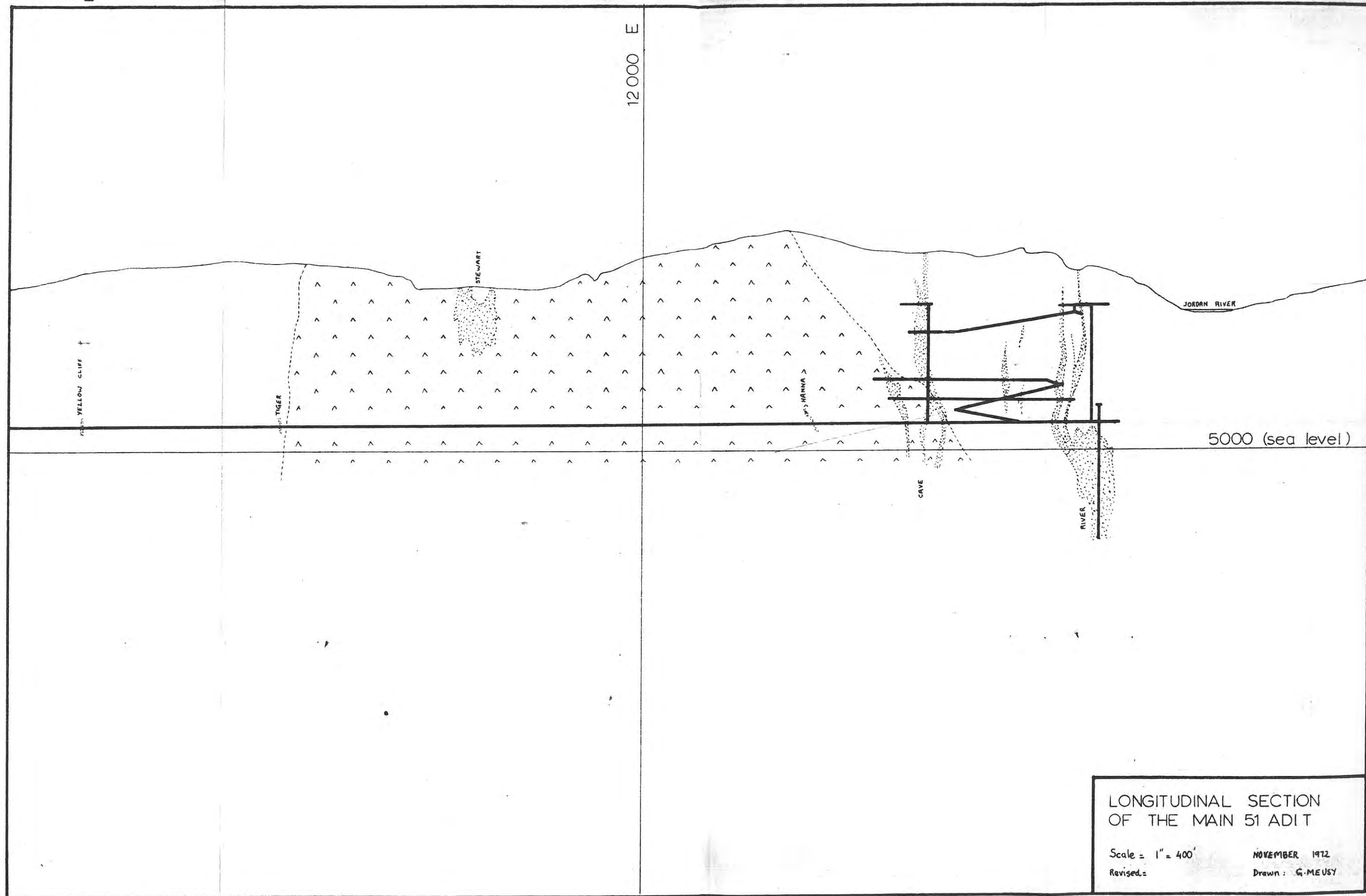




EXP II.2

JORDAN RIVER MINES LTD.
GABBRO MINE JORDAN RIVER, B.C.
CAULFIELD, TIGER, YELLOW CLIFF
GEOLOGY & ASSAYS

DRAWN G. MEUSY	SCALE 1" = 100'
APPROVED	DATE DEC 4, 1971



LONGITUDINAL SECTION
OF THE MAIN 51 ADIT

Scale = 1" = 400'
Revised:

NOVEMBER 1972
Drawn: G. MEUSY